

Fine structure of rupture set for semilinear elliptic equation with singular nonlinearity

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Main Model

The semilinear elliptic equation with *singular nonlinearity* is

$$\Delta u = u^{-p} + f, \quad u \geq 0 \quad \text{in } \Omega \subset \mathbb{R}^n, \quad n \geq 2, \quad p \geq 0, \quad (\text{SN})$$

where Ω is a domain and $f \in L^1_{\text{loc}}(\Omega)$. Some mathematical and physical backgrounds are as follows.

- $p = 0$: $\Delta u = \chi_{\{u>0\}}$, obstacle problem, and free boundary problem.
- $p > 0$:
 - $p > 1$: Thin film theory.
 - $p = 2$: Simplified micro-electromechanical system (MEMS).
 - $p = 1$: Singular minimal hypersurfaces with symmetry.

Example (Dimension one)

For $p = 2$, $n = 1$, and $f \equiv 0$, an easy derivation of the model is the dynamical analysis of two homopolar charges under the Coulomb force.

Motivation

Consider the regularity theory for solutions of (SN) with $p > 0$. If $u \in C^0$ and $f \in C^\infty$, then

$$x \in \{u > 0\} \Rightarrow \exists r > 0, \text{ s.t. } \inf_{B_r(x)} u > 0 \Rightarrow u \in C^\infty(B_r(x)).$$

As a result, the nodal set $\mathcal{N}(u) := \{u = 0\}$ represents the singular set of u , and $\{u > 0\}$ is the regular set. For the thin film theory, $\mathcal{N}(u)$ corresponds to the rupture phenomenon.

Question

How large are rupture sets for solutions of (SN) with $p > 0$? What assumptions will we have to obtain such results?

For $0 < p < 1$, solutions behave like those for the case $p = 0$. The results for the two phases are almost complete (Tavares and Terracini, JMPA 2019). We mainly consider the case $p > 1$, which is very different from $0 < p < 1$.

Problem Settings: Different Solutions

For (SN) with $p > 1$, some definitions are as below.

- **Weak solution:** $u \in (H^1_{\text{loc}} \cap L^{-p}_{\text{loc}})(\Omega)$, and (SN) is satisfied in the sense of distribution.
- **Finite energy solution:** $u \in (C^0_{\text{loc}} \cap H^1_{\text{loc}} \cap L^{1-p}_{\text{loc}})(\Omega)$, and we have (SN) in $\{u > 0\}$, in the sense of distribution.
- **Stationary solution:** weak solution+stationary condition, namely, $\forall Y \in C^\infty_0(\Omega, \mathbb{R}^n)$,

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{F}_f(u(\cdot + tY(\cdot)), \Omega) = 0,$$

where

$$\mathcal{F}_f(u, \Omega) := \int_\Omega \left(\frac{|\nabla u|^2}{2} - \frac{u^{1-p}}{p-1} + fu \right).$$

Previous Results and Remarks

Let d_u be the *Hausdorff dimension* of the rupture set $\mathcal{N}(u)$.

- **Weak solution:**
 - $d_u \leq n - 2 + \frac{4}{p+2}$ (Jiang-Lin, CAM 2004).
 - $d_u \leq n - 2 + \frac{2}{p+1}$ (Dugan-Ponce-Porretta, JAM 2006).
- **Finite energy solution:**
 - $d_u \leq n - 2 + \frac{4}{p+1}$ (Guo-Wei, CPAA 2008).
 - $d_u \leq n - 2 + \frac{2}{p+1}$ (Dávila-Ponce, CRMAS 2008).
- **Stationary solution:**
 - $d_u \leq n - 2 + \frac{4}{p+1}$ (Guo-Wei, MM 2006).
 - $d_u \leq n - 2$ (Dávila-Wang-Wei, AIHP 2016).

The best estimate for d_u is achieved for stationary solutions!

Theorem (Dávila-Wang-Wei, AIHP 2016)

Assume that $u \in C^{0, \frac{2}{p+1}}_{\text{loc}}$ is a stationary solution of $\Delta u = u^{-p}$. Then, $\mathcal{N}(u)$ is a relatively closed set with $d_u \leq n - 2$. In particular, when $n = 2$, the rupture set is discrete.

Remarks:

- The assumption of $\frac{2}{p+1}$ -Hölder continuity is optimal and corresponds to the $C^{1,1}_{\text{loc}}$ regularity in the obstacle problem.
- It is an a priori result. It is still unknown if a stationary solution must be $\frac{2}{p+1}$ -Hölder continuous.
- The result is sharp. If $n = 2$,

$$u(x) = u(|x|) := \left(\frac{2}{p+1} \right)^{-\frac{2}{p+1}} |x|^{\frac{2}{p+1}}$$

is a stationary solution for $\Delta u = u^{-p}$ and $\mathcal{N}(u) = \{0\}$.

- The proof of this result is from the well-known Federer's dimension reduction principle. Similar arguments were applied by (Schoen-Uhlenbeck, JDG 1982) to harmonic maps.

Question (Fine structure of rupture sets)

Can we obtain *further information* about the *stationary solutions* of (SN)? For example, are there properties concerning the regularity of the rupture set?

Main Results

Our results provide a complete characterization for the fine structure of rupture sets of stationary solutions of (SN).

Theorem (Wang-Zhang, arXiv: 2411.16048)

Let $\frac{1}{2} + \frac{1}{2p} < \frac{q}{n}$. $u \in C^{0, \frac{2}{p+1}}_{\text{loc}}(B_4)$ is a stationary solution of (SN) with $f \in L^q_{\text{loc}}(B_4)$, satisfying

$$\|u\|_{L^1(B_2)} + \|f\|_{L^q(B_2)} \leq \Lambda.$$

The following properties hold.

- $\exists(\varepsilon, C) = (\varepsilon, C)(\Lambda, n, p, q) > 0$, s.t.
$$\mathcal{L}^n(B_r(\{u < \varepsilon r^{\frac{2}{p+1}}\} \cap B_1)) \leq Cr^2, \quad \forall 0 < r < 1.$$
- If $f \in W^{j-1, \infty}_{\text{loc}}(B_4)$ with $j \in \mathbb{Z}_+$ and
$$\|f\|_{W^{j-1, \infty}(B_2)} \leq \Lambda',$$
then $D^j u$ has the improved estimate in Lorentz space as
$$\sup \left\{ \lambda > 0 : \lambda^{\frac{2(p+1)}{j(p+1)-2}} \mathcal{L}^n(\{x \in B_1 : |D^j u(x)| > \lambda\}) \right\} \leq C',$$
where $C' = C'(\Lambda, \Lambda', j, n, p, q) > 0$.
- $\mathcal{N}(u)$ is $(n-2)$ -rectifiable, which means that we can cover it with countable numbers of graphs of Lipschitz maps from $\mathbb{R}^{n-2}$ to $\mathbb{R}^n$. For $n = 2$, $\mathcal{N}(u)$ is a discrete set.

Some Remarks on Our Results

- The first point estimates the $(n-2)$ -dimensional *Minkowski r -content* for $\{u < \varepsilon r^{\frac{2}{p+1}}\}$. In particular,
$$\dim_{\text{Min}}(\mathcal{N}(u) \cap B_1) \leq n - 2.$$
- The $L^{\frac{2(p+1)}{j(p+1)-2}, \infty}$ estimate and $(n-2)$ -rectifiability in our main theorem are both sharp.
- By standard interpolations,

$$D^j u \in L^{\frac{2(p+1)}{j(p+1)-2}-}(B_1), \quad \forall j \in \mathbb{Z}_+,$$

i.e. $\forall 0 < s < \frac{2(p+1)}{j(p+1)-2}$, $\exists C = C(\Lambda, \Lambda', j, n, p, q, s) > 0$, s.t.

$$\|D^j u\|_{L^s(B_1)} \leq C.$$

For $j = 1$, $u \in W^{1, \frac{2(p+1)}{p-1}}(B_1)$, improving the H^1 regularity.

- In fact, any k -stratum of $\mathcal{N}(u)$ is k -rectifiable with $k \in \mathbb{Z} \cap [0, n-2]$. The stratification is via the tangent function.
- As an application of our results in the static case of the equation, for a class of suitable weak solutions to the three-dimensional evolutionary problem

$$\partial_t u = \Delta u - u^{-p}, \quad u \geq 0 \quad \text{in } (\Omega \subset \mathbb{R}^3) \times (0, T),$$

where $p > 3$ and $T > 0$, we have that $\{u(\cdot, t) = 0\}$ is 1-rectifiable for a.e. $t \in (0, T)$.

Main Ingredients of the Proof

For simplicity, we consider the simplified case with $f \equiv 0$ in (SN).

1. Refined blow-ups. For $x \in B_2$, let

$$u_{x,r}(y) := \frac{u(x + ry) - u(x)}{r^{\frac{2}{p+1}}}.$$

- If $x \in \mathcal{N}(u)$, then $\exists r_i \rightarrow 0^+$, s.t.
$$u_{x,r_i} \rightarrow u_\infty \text{ strongly in } H^1_{\text{loc}} \cap L^\infty_{\text{loc}} \cap L^{-p}_{\text{loc}},$$
where $\Delta u_\infty = u_\infty^{-p}$ is a stationary solution.
- If $x \notin \mathcal{N}(u)$, then as $r \rightarrow 0^+$,
$$u_{x,r} \rightarrow 0 \text{ strongly in } H^1_{\text{loc}} \cap L^\infty_{\text{loc}}.$$

Definition (Stratification)

$\forall k \in \mathbb{Z} \cap [0, n-1]$, define the k -stratum of u by

$$S^k(u) := \{x \in B_2 : \text{no tangent function } v \text{ of } u \text{ at } x \text{ is } (k+1)\text{-symmetric}\}.$$

Then, $S^0(u) \subset S^1(u) \subset S^2(u) \subset \dots \subset S^{n-2}(u) = B_2$.

2. Quantitative stratification. These concepts were introduced by (Cheeger-Naber IM. 2013) first in the study of Gromov-Hausdorff limits. (Naber and Valtorta, AM 2017) developed Reifenberg-rectifiable methods to improve it.

Definition (Quantitative symmetry)

Let $\varepsilon > 0$ and $k \in \mathbb{Z} \cap [0, n]$. u is (k, ε) -symmetric in $B_r(x)$, if there is a k -symmetric function $h \in C^{0, \frac{2}{p+1}}_{\text{loc}}(\mathbb{R}^n)$ s.t.

$$\|u_{x,r} - h\|_{L^\infty(\bar{B}_1)} < \varepsilon.$$

3. Dichotomy result. We will deal with points in B_2 through a dichotomy result.

- $0 \leq u(x) \ll r^{\frac{2}{p+1}}$:
The behavior of u within $B_r(x)$ is like the case that $u(x) = 0$. Thus, we apply methods by Naber and Valtorta.
- $u(x) \gtrsim r^{\frac{2}{p+1}}$:
We use the standard regularity theory for elliptic equations to find a small ball $B_{\delta r}(x)$ with $\delta \ll 1$ s.t. u exhibits nice properties in it.

^a k -symmetry refers to functions that are $\frac{2}{p+1}$ -homogeneous and invariant for k directions.